
Python Environment Setup

What, if any, constraints are there on the Python setup for this project?

How to set up the project from the CLI?

How to set up the project using PyCharm?

What are the required packages that need to be installed?

What is the HF_TOKEN and how do I get it and use it?

What, if any, constraints are there on the Python setup?

- The **Langchain** framework is made up of many packages
- Langchain **connects LLMs to Private Data sources** used in RAG
- The **FAISS index**, which acts like a database of Private Data is in the **langchain_community** package.
- **Python 3.13 is the recommended version for Langchain**



Use
Python
version
3.13

Code Setup with CLI: [See RAG_313 README.md](#)

1. Install Python version [3.13](#).
2. Download the [RAG_313](#) repository and it will be automatically named **RAG_313-main**.
3. Change directory into **RAG_313-main**. Create a virtual environment:
python3.13 -m venv .venv.
4. Activate the virtual environment:
 - a. **source .venv/bin/activate** (MAC/LINUX)
 - b. **.venv\Scripts\activate** (WINDOWS COMMAND PROMPT)
5. The directory will look like this: **(.venv) rag_313-main>**
6. Install packages: **pip install -r requirements.txt**
7. You can end the virtual environment using **deactivate**

How to set up the project using PyCharm?

- Read the **PYCHARM.md** document
- Follow instructions for installing Python v3.13 and packages

What are the required packages that need to be installed?

convert.py

docling

ingest.py

langchain_community
langchain_openai
langchain_text_splitters
langchain_core

app.py

langchain_openai
langchain_community
langchain_core
sentence_transformers
langchain_core
transformers

see [requirements.txt](#)

What is the HF_TOKEN and how do I get it and use it?

- **sentence_transformers** package installs a model locally that will be used for ranking the retrieved user data with respect to its similarity with the contents of the user query
 - **Linux/macOS:** `~/.cache/torch/sentence_transformers`
 - **Windows:** `<username>\.cache\torch\sentence_transformers`
- **add an HF_TOKEN to a .env file to avoid raising an authentication error**
 - HF_TOKEN is created on <https://huggingface.co/>
 - create a free account on huggingface.co
 - go to settings/click on access tokens
 - create a new token/give it a name/make it read only
 - generate a new token and copy it into .env
 - Your entry in .env will look like this where the token is 37-40 characters:
HF_TOKEN="hf_abcdefghijklmnopqrstuvwxyz0123456789"

Create a free account and click on profile

The screenshot shows the Hugging Face website interface. At the top left is the Hugging Face logo. Next to it is a search bar with the placeholder text "Search models, datasets, users...". To the right of the search bar are navigation links: Models, Datasets, Spaces, Buckets, Docs, and Pricing. A green arrow points to a profile icon in the top right corner, which is a circular button with a purple and blue gradient. Below the navigation links, the main content area is divided into three columns. The left column contains a sidebar with a "New" button, a user profile for "rebeccapeltz" with links to Profile, Inbox (0), Settings, Billing, and Get PRO, and sections for Organizations and Resources. The middle column features a "Following" section with tabs for All, Models, Datasets, Spaces, Buckets, Papers, Collections, and Community. Below this is a card titled "Spaces agents.md for your coding agents" with a description and a "Hugging Face Changelog" button. At the bottom of the middle column is a "Follow your favorite AI creators" section with buttons to follow "Etched", "PaddlePaddle", and "IndexTeam". The right column displays a "Trending" section with a list of models and datasets, including "SulphurAI/Sulphur-2-base", "openbmb/MiniCPM-V-4.6", "Supertone/supertonic-3", "bytedance-research/Lance", "PsiBotAI/SynData", "AlienKevin/SWE-ZERO-12M-trajectories", "angrygiraffe/claude-opus-4.6-4.7-reaso...", and "TuringEnterprises/Open-MM-RL".

Hugging Face

Search models, datasets, users...

Models Datasets Spaces Buckets NEW Docs Pricing

+ New

rebeccapeltz

- Profile
- Inbox (0)
- Settings
- Billing
- Get PRO

Organizations

- Create New

Resources

- Getting Started
- Documentation
- Forum
- Tasks
- Learn

Light theme

Following 0

All Models Datasets Spaces Buckets Papers Collections Community

Posts Upvotes Likes Articles

Spaces agents.md for your coding agents

Every Gradio Space now auto-serves an `/agents.md` endpoint, a machine-readable API description that AI agents can read and call directly. Point your coding agents (like Claude Code,...

Hugging Face Changelog

296 Apr 17

NEW Follow your favorite AI creators Refresh List

- Etched · Integrating generative AI with gaming experiences Follow
- PaddlePaddle · Document intelligence in compact vision models Follow
- IndexTeam · Expressive TTS and anime video generation Follow

Trending last 7 days

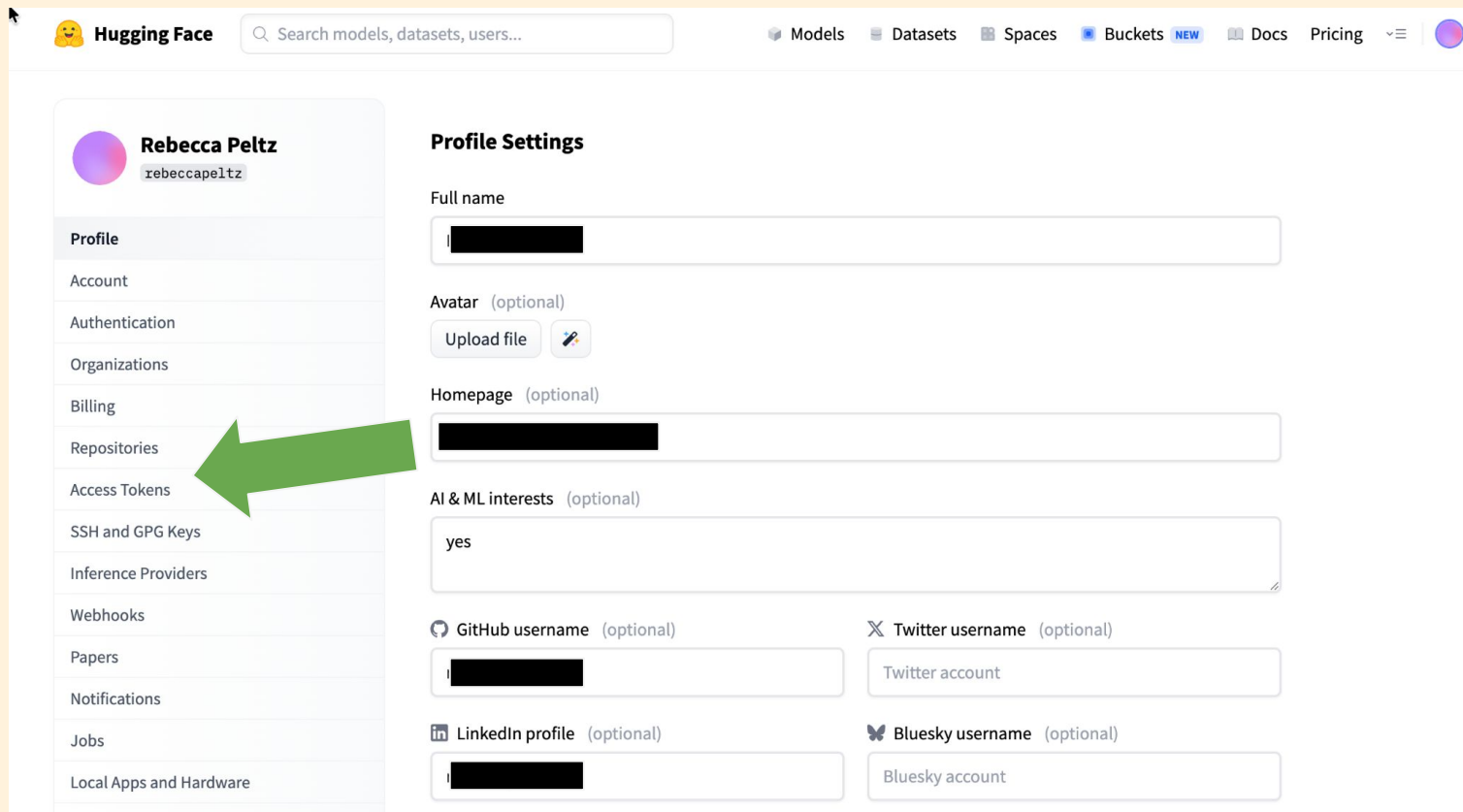
All Models Datasets

- SulphurAI/Sulphur-2-base
Text-to-Video · 9B · Update... · 1.11M · 1.17k
- openbmb/MiniCPM-V-4.6
Image-Text-to-Text · 1B · Upd... · 145k · 802
- Supertone/supertonic-3
Text-to-Speech · Updated 1 day a... · 28.7k · 462
- bytedance-research/Lance
Any-to-Any · Updated about 4 hou... · 171 · 289
- PsiBotAI/SynData
Viewer · Updated abou... · 449k · 35.7k · 145
- AlienKevin/SWE-ZERO-12M-trajectories
Viewer · Updated 5 day... · 12.3M · 7.57k · 81
- angrygiraffe/claude-opus-4.6-4.7-reaso...
Viewer · Updated 18 d... · 38.5k · 3.17k · 145
- TuringEnterprises/Open-MM-RL
Viewer · Updated 7 days... · 40 · 8.17k · 120

Next Click on Settings

The screenshot shows the Hugging Face website interface. On the left sidebar, the user's profile 'rebeccapeltz' is listed with links to Profile, Inbox (0), Settings, Billing, and Get PRO. Below this are sections for Organizations and Resources. The main content area features a 'Following' section with tabs for Models, Datasets, Spaces, Buckets, Papers, Collections, and Community. A post titled 'Spaces agents.md for your coding agents' is visible. On the right, a 'Trending' section lists various models. A profile dropdown menu is open for 'rebeccapeltz', showing options like Notifications, New Model, New Dataset, New Space, New Bucket, New Collection, Create organization, Usage Quota, Private Storage, Public Storage, Zero GPU, Inference Usage, Get Hugging Face PRO, **Settings**, and Access Tokens. A large green arrow points from the 'Settings' link in the dropdown menu to the 'Settings' link in the left sidebar.

Next Click Access Tokens



The screenshot shows the Hugging Face user interface. At the top, there's a navigation bar with the Hugging Face logo, a search bar, and links for Models, Datasets, Spaces, Buckets (marked as NEW), Docs, and Pricing. The left sidebar displays the user's profile information and a list of settings: Profile, Account, Authentication, Organizations, Billing, Repositories, Access Tokens, SSH and GPG Keys, Inference Providers, Webhooks, Papers, Notifications, Jobs, and Local Apps and Hardware. A large green arrow points to the 'Access Tokens' link. The main content area is titled 'Profile Settings' and contains several form fields: Full name, Avatar (optional) with an 'Upload file' button, Homepage (optional), AI & ML interests (optional) with a text area containing 'yes', GitHub username (optional), Twitter username (optional), LinkedIn profile (optional), and Bluesky username (optional). Each of these fields has a corresponding input box, some of which are redacted with black boxes.

Hugging Face Search models, datasets, users...

Models Datasets Spaces Buckets **NEW** Docs Pricing

Rebecca Peltz
rebeccapeltz

Profile
Account
Authentication
Organizations
Billing
Repositories
Access Tokens
SSH and GPG Keys
Inference Providers
Webhooks
Papers
Notifications
Jobs
Local Apps and Hardware

Profile Settings

Full name
[Redacted]

Avatar (optional)
Upload file [Icon]

Homepage (optional)
[Redacted]

AI & ML interests (optional)
yes

GitHub username (optional)
[Redacted]

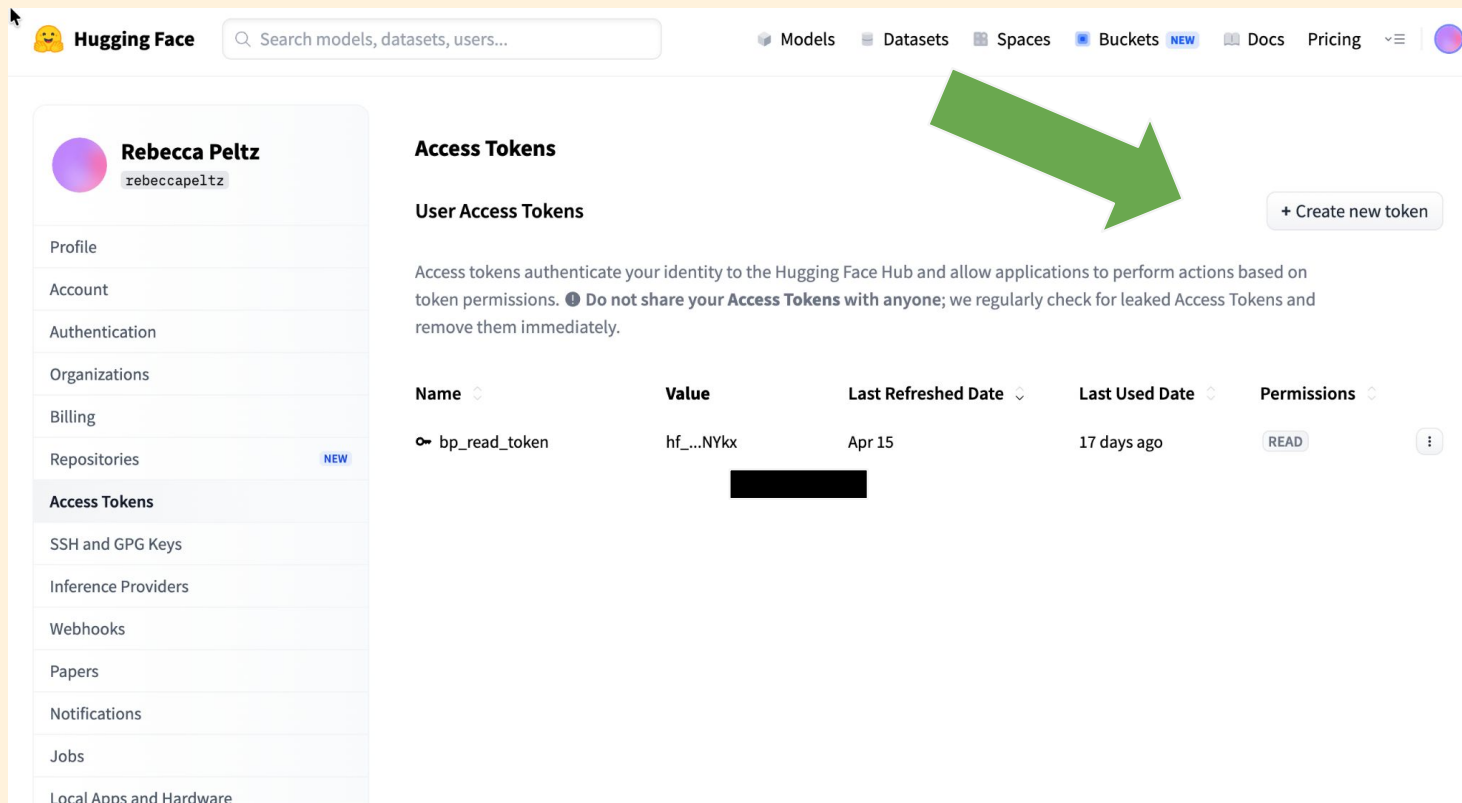
Twitter username (optional)
Twitter account

LinkedIn profile (optional)
[Redacted]

Bluesky username (optional)
Bluesky account

Create a new READ token and give it a name

A “Value” will be created



The screenshot shows the Hugging Face user profile for Rebecca Peltz. The left sidebar contains navigation links: Profile, Account, Authentication, Organizations, Billing, Repositories (marked as NEW), Access Tokens (highlighted), SSH and GPG Keys, Inference Providers, Webhooks, Papers, Notifications, Jobs, and Local Apps and Hardware. The main content area is titled 'Access Tokens' and includes a sub-section 'User Access Tokens'. A green arrow points to the '+ Create new token' button. Below this, a table lists existing tokens. The first token is named 'bp_read_token' with a value 'hf_...NYkx' (partially redacted with a black box), last refreshed on 'Apr 15', last used '17 days ago', and has 'READ' permissions. A warning message states: 'Access tokens authenticate your identity to the Hugging Face Hub and allow applications to perform actions based on token permissions. Do not share your Access Tokens with anyone; we regularly check for leaked Access Tokens and remove them immediately.'

Name	Value	Last Refreshed Date	Last Used Date	Permissions
bp_read_token	hf_...NYkx	Apr 15	17 days ago	READ

Add the “Value” to your local **.env** file

The value will be 36-40 characters.

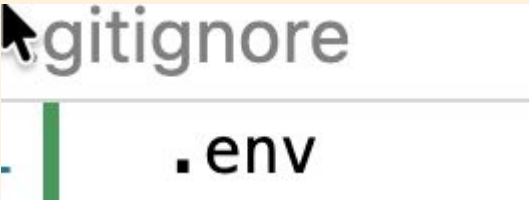
If you’re uploading to GitHub, add **.env** to **.gitignore**



.env

A screenshot of a file explorer window. The file ".env" is selected, and its contents are displayed in a text area below. The text area has a light blue background. The content is "HF_TOKEN=" followed by a long string of red characters: "ABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789".

```
HF_TOKEN="ABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789"
```



.gitignore

A screenshot of a file explorer window. The file ".gitignore" is selected, and its contents are displayed in a text area below. The text area has a light blue background. The content is ".env".

```
.env
```

Docker Model Runner Setup

What is Docker and how to install it on your local computer?

What is Docker and Docker Model Runner?

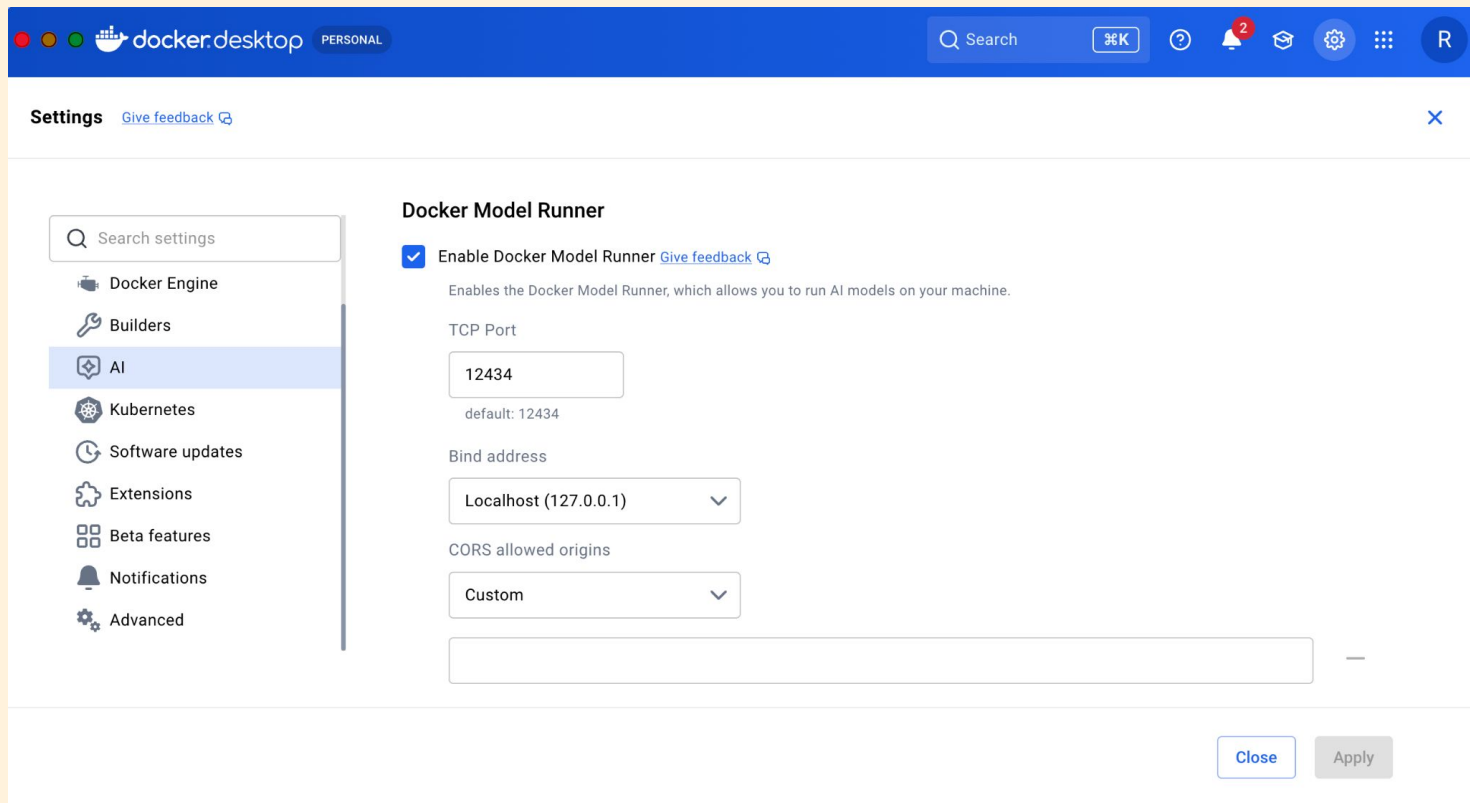
How to enable Docker Model Runner?

How to install models, list them and test them?

What is Docker and Docker Model Runner?

- **Docker** is
 - open source platform
 - used to build, test, and deploy software
 - uses containers
- **Docker Model Runner** is a feature that allows you to
 - download, manage and execute open source large language models

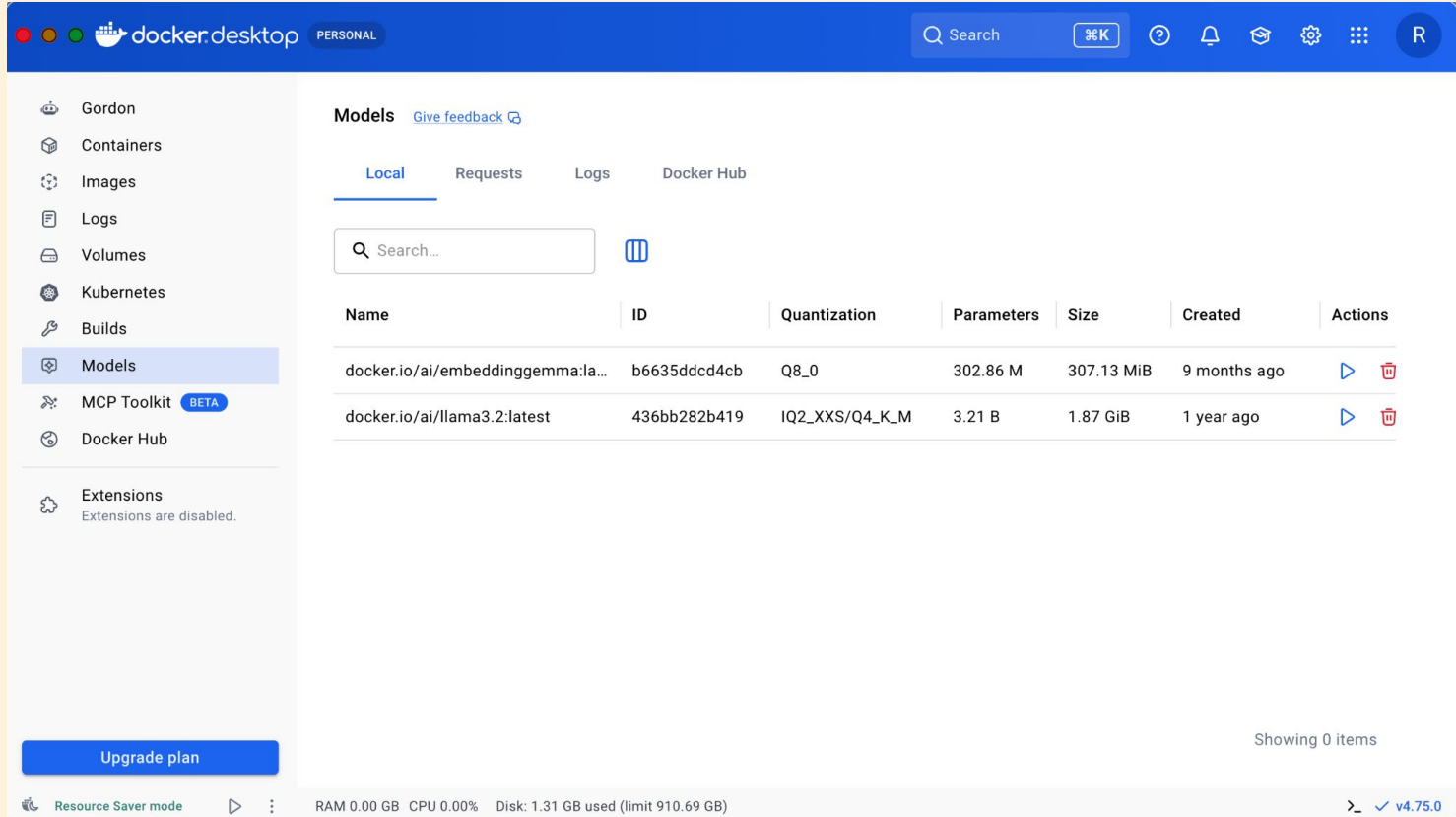
Enable AI and TCP with the User Interface



Manage Models with CLI

```
# Pull models  
# Model used to create FAISS Index/Ingestion Pipeline  
docker model pull ai/embeddinggemma  
  
# LLM used for Query Pipeline  
docker model pull ai/llama3.2  
  
# List Downloaded Models  
docker model list  
  
# Run the Chat Model  
docker model run ai/llama3.2  
  
# End Chat Model Conversation  
/bye  
  
# Run the Embeddings Model  
# curl will make an API call to convert text to a numeric vector  
curl --location 'http://localhost:12434/engines/llama.cpp/v1/embeddings' \  
--header 'Content-Type: application/json' \  
--data '{ "model": "ai/embeddinggemma", "input": "Your text to embed here" }'
```

Run and Delete Models from the UI



The screenshot shows the Docker Desktop interface with the 'Models' tab selected. The left sidebar contains navigation options: Gordon, Containers, Images, Logs, Volumes, Kubernetes, Builds, Models (highlighted), MCP Toolkit (BETA), Docker Hub, and Extensions (disabled). The main area displays the 'Models' section with tabs for Local, Requests, Logs, and Docker Hub. A search bar is present. Below it is a table of models:

Name	ID	Quantization	Parameters	Size	Created	Actions
docker.io/ai/embeddinggemma:la...	b6635ddcd4cb	Q8_0	302.86 M	307.13 MiB	9 months ago	 
docker.io/ai/llama3.2:latest	436bb282b419	IQ2_XXS/Q4_K_M	3.21 B	1.87 GiB	1 year ago	 

At the bottom right of the table area, it says 'Showing 0 items'. The bottom status bar shows 'Resource Saver mode', system resource usage (RAM 0.00 GB, CPU 0.00%, Disk 1.31 GB used), and the version 'v4.75.0'.

Docker Model Runner Setup Demo

If you have worked along with the demo you should be able to interact with the models on your local computer using the Command Line Interface or Docker User Interface.

```
# Pull models
# Model used to create FAISS Index/Ingestion Pipeline
docker model pull ai/embeddinggemma

# LLM used for Query Pipeline
docker model pull ai/llama3.2

# List Downloaded Models
docker model list

# Run the Chat Model
docker model run ai/llama3.2

# End Chat Model Conversation
/bye

# Run the Embeddings Model
# curl will make an API call to convert text to a numeric vector
curl --location 'http://localhost:12434/engines/llama.cpp/v1/embeddings' \
--header 'Content-Type: application/json' \
--data '{ "model": "ai/embeddinggemma", "input": "Your text to embed here" }'
```

LLM Setup with Docker Model Runner

1. [Download Docker Desktop](#) You can use Docker Personal Plan for free.
2. Set up Docker to load and run the two models: **ai/llama3.2** and **ai/embeddinggemma**
3. Use the CLI (Command Line Interface) to load the models:
 - a.
4. After conversion and ingestion, run the app: `python app.py`
5. (Optional) If you're using this to learn how the RAG flow behaves, you can run the `app_debug.py` script to get information back at each step.

```
docker desktop enable model-runner
```

```
docker desktop enable model-runner --tcp 12434 (host processes and external apps)
```

```
docker model pull ai/embeddinggemma
```

Python Virtual Environment Setup Demo

If you have worked along with the demo you should have the rag_313 code downloaded. The code should be using Python 3.13 as an interpreter, there should be a virtual environment with all of the packages in requirements.txt installed. Calling pip check should let you know that there are no broken packages.

```
(.venv) becky@Rebeccas-MacBook-Pro-2 rag_313-main % pip check  
No broken requirements found.  
(.venv) becky@Rebeccas-MacBook-Pro-2 rag_313-main %
```